BANGLADESH UNIVERSITY OF PROFESSIONALS

Department of Computer Science and Engineering

Faculty of Science and Technology

Software Development Project-I Laboratory

CSE-3208

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Tuition Track Management System Project Proposal

1. Title of the Project

Tuition Track Management System – A web-based system for managing tuition classes, student records, payments, and schedules.

2. Executive Summary

The Tuition Track Management System is a web application designed to help tuition teachers and coaching centers manage students, classes, attendance, and payments efficiently. Many teachers currently manage tuition records manually which can lead to data loss, difficulty tracking payments, and disorganized schedules.

• Centralized system for managing students
• Automated payment tracking
• Organized class schedules
• Improved communication between teachers and students

3. Project Objectives

- **To develop a system for** managing student information **in a structured and digital way**
- **To** record and track student attendance **efficiently**
- **To** monitor tuition fee payments **and maintain payment history**
- **To** organize and manage class schedules **without conflicts**
- **To provide a** secure login system **for authorized users**
- **To** generate useful reports **such as attendance and payment summaries**
- **To** send notifications and reminders **for fees and classes**
- **To** reduce manual work and improve overall efficiency **of tuition management**

4. Scope of Work

- **User Management:** Registration and login system for teachers/admin and students
- **Student Management:** Add, update, delete, and view student information
- **Attendance System:** Record and maintain daily attendance of students
- **Fee Management:** Track tuition fees, payment status, and payment history
- **Class Scheduling:** Create and manage class routines and timings
- **Dashboard:** Provide an overview of students, attendance, and payments
- **Reporting:** Generate reports for attendance and fee records
- **Notification System:** Send reminders for classes and fee payments
- **Data Storage:** Maintain a secure database for all records
- **System Access:** Accessible via web browsers on desktop and mobile devices

Deliverables:

- Fully functional web application
- Database system
- User interface (UI)
- Documentation

5. Target Audience

The Tuition Management System is designed for the following users:

- **Tuition Teachers:**
  To manage student records, attendance, schedules, and fee tracking بسهولة
- **Coaching Centers or Institutes:**
  To handle multiple students, batches, and overall administration efficiently
- **Students:**
  To check class schedules, receive notifications, and stay updated on fees
- **Administrators:**
  To monitor the entire system, manage users, and generate reports

This system is mainly intended for anyone involved in managing or attending tuition

6. Technology Stack

The project involves developing a **modern web-based tuition management system** using **Hono, React, PostgreSQL, and Bun** for fast and efficient performance

1. Frontend Development (React)

- Build a responsive and user-friendly interface using **React**
- Create pages for dashboard, student management, attendance, payments, and schedules
- Implement forms, tables, and navigation components

2. Backend Development (Hono + Bun)

- Develop REST APIs using **Hono** running on **Bun**
- Handle business logic for student data, attendance, and payments
- Implement authentication and authorization system

3. Database Management (PostgreSQL)

- Design and manage relational database using **PostgreSQL**
- Store student records, attendance, fees, and schedules
- Ensure data consistency and efficient querying

4. Core Features Implementation

- Student registration and profile management
- Attendance tracking system
- Fee/payment tracking and history
- Class scheduling and management
- Dashboard with summary data

5. Authentication & Security

- Secure login system (JWT/session-based)
- Role-based access (admin, teacher, student)
- Data protection and validation

6. API Integration

- Connect frontend (React) with backend (Hono APIs)
- Handle data fetching and state management

7. Testing & Debugging

- Test system functionality (frontend + backend)
- Fix bugs and ensure smooth performance

8. Deployment

- Deploy backend using Bun-supported environment
- Host frontend and connect with backend APIs
- Configure database hosting (PostgreSQL cloud/local)

9. Documentation

- Provide system documentation
- Include API documentation and user guide

This scope ensures a **full-stack, scalable, and high-performance system** using modern technologies.

7. Project Timeline

The project will be completed in 6 phases over the course of the semester:

Phase	Task	Duration
1	Requirement analysis	Week 1–2
2	UI/UX system design	Week 3–4
3	Frontend development (React)	Week 5–7
4	Backend development (Hono & Bun)	Week 8–10
5	Testing & debugging	Week 11–12
6	Deployment & documentation	Week 13–14

The timeline may be adjusted based on project progress and course requirements.

8. Team Structure

The tuition track project will be developed by a team of 3 students. Each member is responsible for specific areas of the project, though all members will collaborate and support each other throughout the development process.

Member	Role	Responsibility
Faiza	Frontend Developer	Plans the project, manages the timeline, designs UI/UX using Figma, and develops the interface using **React, HTML, CSS, and JavaScript**.
Tuly	Backend Developer	Develops server-side logic using **Hono with Bun**, builds REST APIs, handles authentication, and manages attendance, fee, and schedule functionalities.
Alvee	Database Administrator & Full Stack Support	Designs and manages the **PostgreSQL database**, integrates it with the backend, and supports features like payment tracking, reports, and system optimization.

9. Success Metrics

The success of the Tuition Management System will be evaluated using the following metrics:

- **System Usability:** Users can easily navigate and use the system without difficulty.
- **Efficiency Improvement:** Reduction in manual record keeping for attendance, fees, and student data.
- **Accuracy of Data:** Student records, attendance, and payment information are stored and managed correctly.

- **System Performance:** The system runs smoothly with fast response time.
- **User Satisfaction:** Teachers and administrators are satisfied with the system's functionality.
- **Feature Functionality:** All core features such as student management, attendance tracking, fee management, and scheduling work properly.

These metrics help determine whether the system meets its objectives and improves tuition management.

10. Maintenance & Support

Ensure smooth operation by fixing bugs, updating software, optimizing performance, backing up data, and providing user/technical support. Monitor system health, apply security patches, and respond to issues based on priority